

AWS Managed Database & Storage Concepts

Managed Database Services in AWS

Managed database services in AWS mean AWS handles provisioning, patching, scaling, backups, high availability, and monitoring. Developers focus only on using the database.

Examples include:

Service	Description	Best Use Cases
Amazon RDS	Relational DB (MySQL, PostgreSQL, Oracle, SQL Server, MariaDB)	SQL Server, MySQL, RDB ecommerce
Amazon Aurora	Cloud-optimized RDS, MySQL/PostgreSQL compatible	High performance relational workloads
Amazon DynamoDB	NoSQL key-value/document DB, serverless	IoT, gaming, ecommerce, real-time apps
Amazon DocumentDB	Managed MongoDB-compatible document DB	JSON/document data
Amazon ElastiCache	In-memory caching (Redis, Memcached)	Performance, session stores
Amazon Neptune	Graph database	Fraud detection, recommendation engines
Amazon Keyspaces	Managed Cassandra	Cassandra workloads
Amazon Timestream	Time-series database	IoT, monitoring
Amazon QLDB	Ledger database with immutable logs	Financial, supply chain

Transactional (OLTP) vs Analytical (OLAP) Databases

Feature	Transactional (OLTP)	Analytical (OLAP)
Purpose	Day-to-day transactions	Complex queries & reporting
Workload	Frequent small read/writes	Heavy read, large aggregations
Data Model	Normalized schema	Star/denormalized schema
Query Type	Simple inserts/updates	Aggregations, summaries
Users	Customers, apps	Data analysts, BI teams
AWS Services	RDS, Aurora, DynamoDB	Redshift, Athena, EMR

Analogy: OLTP is 'running the business' (transactions), while OLAP is 'analyzing the business' (insights).

AWS Data Pipeline

AWS Data Pipeline is a managed ETL service that helps you move, transform, and process data across AWS services and on-premises. It can schedule workflows, retry on failure, and scale automatically.

Common Use Cases: ETL jobs (S3 → EMR → Redshift), data backup (DynamoDB → S3), cross-region transfers, and data analytics pipelines.

Service	Purpose
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AWS Data Pipeline	General ETL/data movement workflows (batch)
AWS Glue	Modern ETL with schema discovery, serverless
Kinesis	Real-time streaming data
Step Functions	Workflow orchestration (apps & data)

Best Ways to Deploy Frontend Apps in AWS

- 1. Amazon S3 + CloudFront → Cheapest & most common, for static sites/SPAs.
 - 2. AWS Amplify → Developer-friendly, built-in CI/CD integration with Git.
 - 3. Elastic Beanstalk → Deploy frontend + backend together.
 - 4. EC2 (manual) → Use Apache/Nginx (not recommended for static apps).
 - 5. Lightsail → VPS-like, simple hosting for small projects.
 - 6. ECS/EKS → Containerized deployment (microservices).
- Recommended: S3 + CloudFront (static hosting + CDN + Route 53).

Object Storage Services in AWS

Object storage stores data as objects (data + metadata + unique ID). It's flat, scalable, and ideal for unstructured data (images, videos, logs, backups).

Type	AWS Service	Best Use Case
Object Storage	Amazon S3, Glacier	Media, backups, data lakes
Block Storage	EBS, Instance Store	Databases, OS disks
File Storage	EFS, FSx	Shared file systems, legacy apps